Set-A

BRAC University **Department of Computer Science and Engineering** Semester: **Summer 2025** Course Code: **CSE460** Course Title: **VLSI Design**	**Final Examination** Full Marks: **15 x 3 = 45** Time: **1 hour 30 minutes** Date: **14 September 2025**

Student ID:	**Name:**	**Section:**

[Answer any **THREE** questions. **Return** the question paper with the answer script.]

<u>Question 1:</u> [CO4]

The graph below can be optimally partitioned using the **KL algorithm**. The dotted line represents the initial partitioning. Assume all the **edges** have the **same weight**.

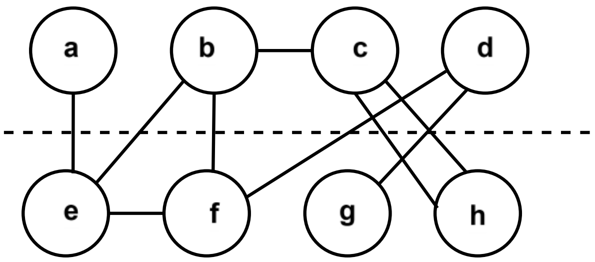


Figure 1

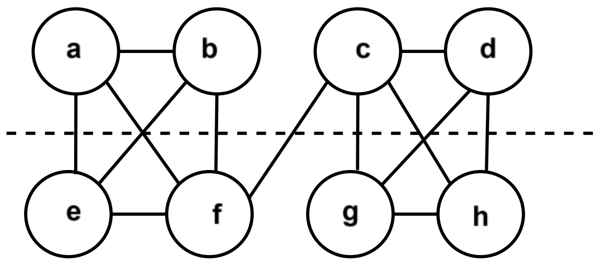


Figure 2

(a)	What are the common steps of **Physical Design**?	[2]
(b)	**Find** the **iteration number** in the first pass of the KL algorithm for **Figure 1**. Calculate the **costs** of moving each node from one partition to another in **Figure 1**.	[2]
(c)	**Perform** the **first iteration** of the KL algorithm for **Figure 1** and **swap** the best pair of nodes. After that, find the old cut cost and the new cut cost.	[7]
(d)	For Figure 2, the **best gain** value of different iterations is given below, $\Delta g_1 = 3$ $\Delta g_2 = 5$ $\Delta g_3 = -6$ $\Delta g_4 = -2$ (i) **Calculate** the **cumulative gain** of every iteration of the first pass. (ii) Are any further passes necessary? **State** the reason for your answer.	[4]

Question 2: [CO4]

For question (a) you are given a certain **chip planning** task. You have **four blocks** with the following areas. The dimension of each small grid is $\mathbf{0.5} \times \mathbf{0.5}$

Block A: $A = 2$

Block B: $A = 9$

Block C: $A = 4$

Block D: $A = 3$

are metals and , are obstacles

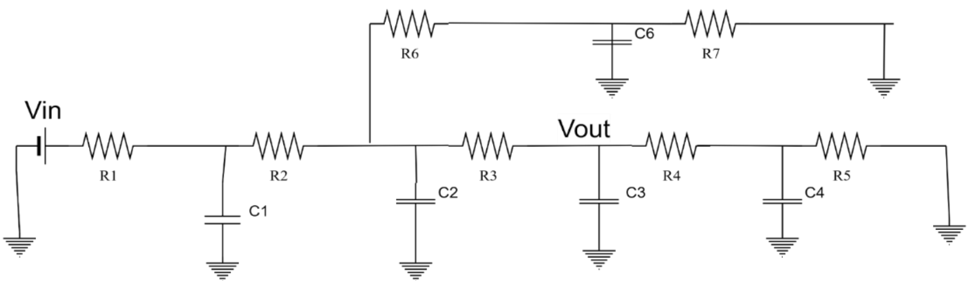
(a)	**Find** and draw a floor plan with the minimum total area enclosed if the **global bounding box** is a $\mathbf{20} \times \mathbf{20}$ grid.	[3]
(b)	**Find** the **Manhattan** Distance from S to T1.	[2]
(c)	Using **Lee's Maze Algorithm**, find the shortest path from S to T1 and T2, allowing minimum bends	[4]
(d)	**Find** the memory requirement for the algorithm if we use a sequence of **natural numbers**. If the **maximum memory** requirement of the provided grid is **45 bytes**, find how many **unique numbers** starting from 0 can be used for wave propagation in **Lee's maze** algorithm.	[6]

Question 3: [CO3]

Part1: The **Texas Instrument** company's analog engineers are designing a **CMOS 3- input NAND gate** considering delay and power consumption. They have chosen GaN semiconductor which has $\boldsymbol{\mu_n = 3\mu_p}$.

They have **fixed** the width scaling factor $\boldsymbol{k_n = 2}$ for **NMOS** transistors and want to meet **equal fall and rise resistance**. Later, they are informed that the **NAND gate** will drive a total load of **three NOT** gates and **two OR** gates.

Part 2: There is a **RC tree** network given below.



(a)	**Find** the **width** scaling factor for **PMOS** ($\boldsymbol{k_p}$) to meet the rise and fall resistance specification.	[2]
(b)	**Draw** the **RC** equivalent circuit of the **NAND gate** considering the loads are not connected. **Use** the **Elmore delay** model to find the expressions for $t_{cdr}, t_{pdr}, t_{cdf}$, and t_{pdf}.	[7]
(c)	If each **NOT gate** and **OR gate** load contributes $\boldsymbol{3C}, \boldsymbol{5C}$ unit capacitance respectively, then after connecting the load estimate how many times slower the **NAND gate** output will operate compared to **load disconnected** condition considering **propagation delay rising** for both cases.	[3]
(d)	**In Part2**, if the R_1 , R_2, R_3 , R_4 and R_6 path is **on** then estimate the **worst case** rising delay using **Elmore delay** model associated with V_{in} to be propagated to V_{out}.	[3]

<u>Question 4:</u> [CO3]

TSMC are planning to design a new chip in a $\mathbf{1.1\ V}$, $\mathbf{22\ nm}$ ($\boldsymbol{\lambda = 10\ nm}$) technology which has $\mathbf{200}$ million transistors, of which $\mathbf{25\%}$ are used for building logic gates and the rest are used for memory arrays. The average **logic transistor** width is **6λ** and the average **memory transistor** width is **4λ**. Other information of the device is given below

i. The gate and diffusion capacitances are $\mathbf{1.2}\ \boldsymbol{fF/\mu m}$ and $\mathbf{0.6}\ \boldsymbol{fF/\mu m}$
ii. The system clock frequency is $\mathbf{3.5}\ \boldsymbol{GHz}$. In every $\mathbf{100}$ $\boldsymbol{clock\ cycles}$, the number of **switching** for the **logic** and **memory** transistors are $\mathbf{10}$ and $\mathbf{2}$ respectively.
iii. For both of the transistors, each transistor contributes the gate leakage, junction leakage and subthreshold leakage currents of approximately $\mathbf{20}\ \boldsymbol{pA}$, $\mathbf{15}\ \boldsymbol{pA}$ **and** $\mathbf{25}\ \boldsymbol{pA}$ respectively. The total short circuit power is $\mathbf{0.05}\ \boldsymbol{W}$. The total accepted power consumption is $\mathbf{3}\ \boldsymbol{W}$.

(a)	**Discuss** in brief how the **short circuit power** and **subthreshold leakage power** dissipate in the device.	[3]
(b)	**Calculate** the **activity factor** and **switching frequency** for each type of transistors.	[2]
(c)	**Calculate** the **dynamic power** and **static power** consumption.	[6]
(d)	**Verify** the statement that “By lowering the device **speed** by 10% we can meet the power consumption limit”.	[4]